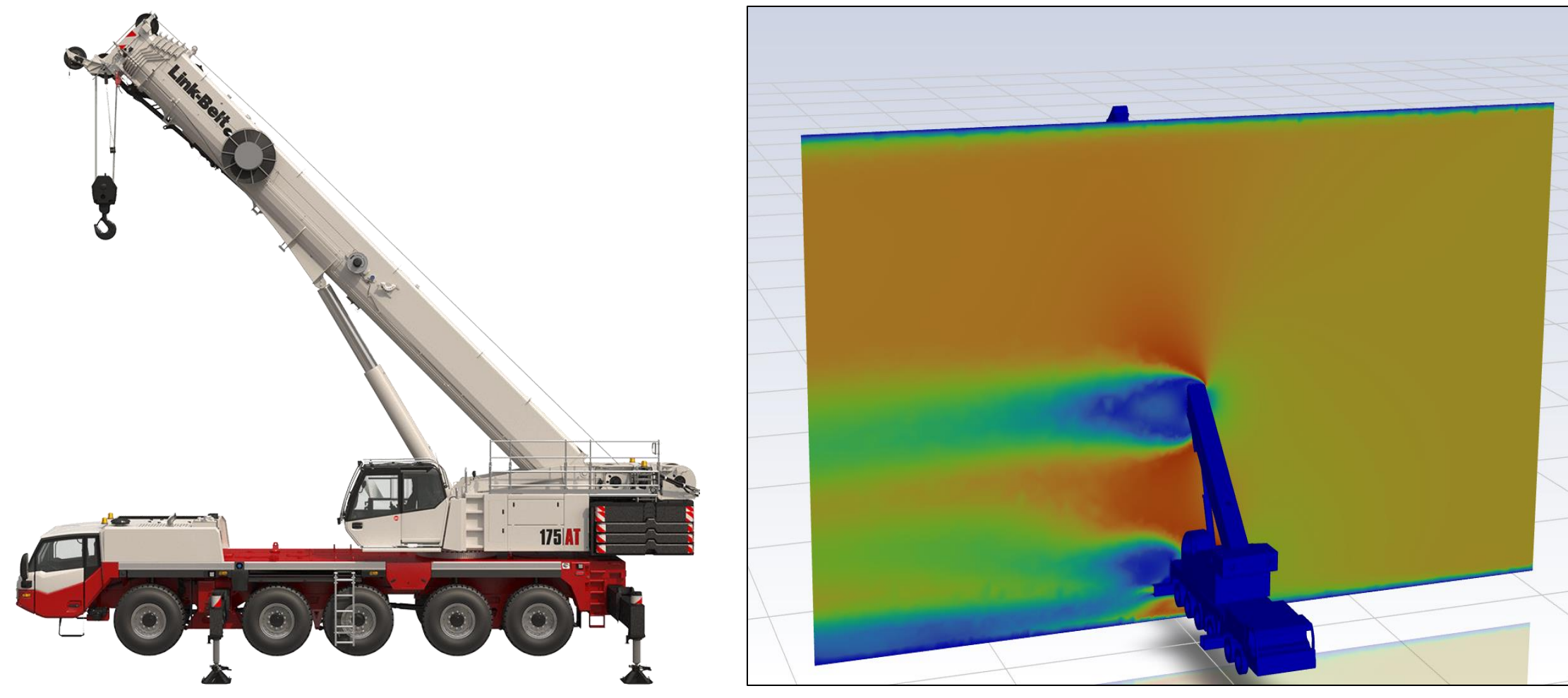


Background & Motivation

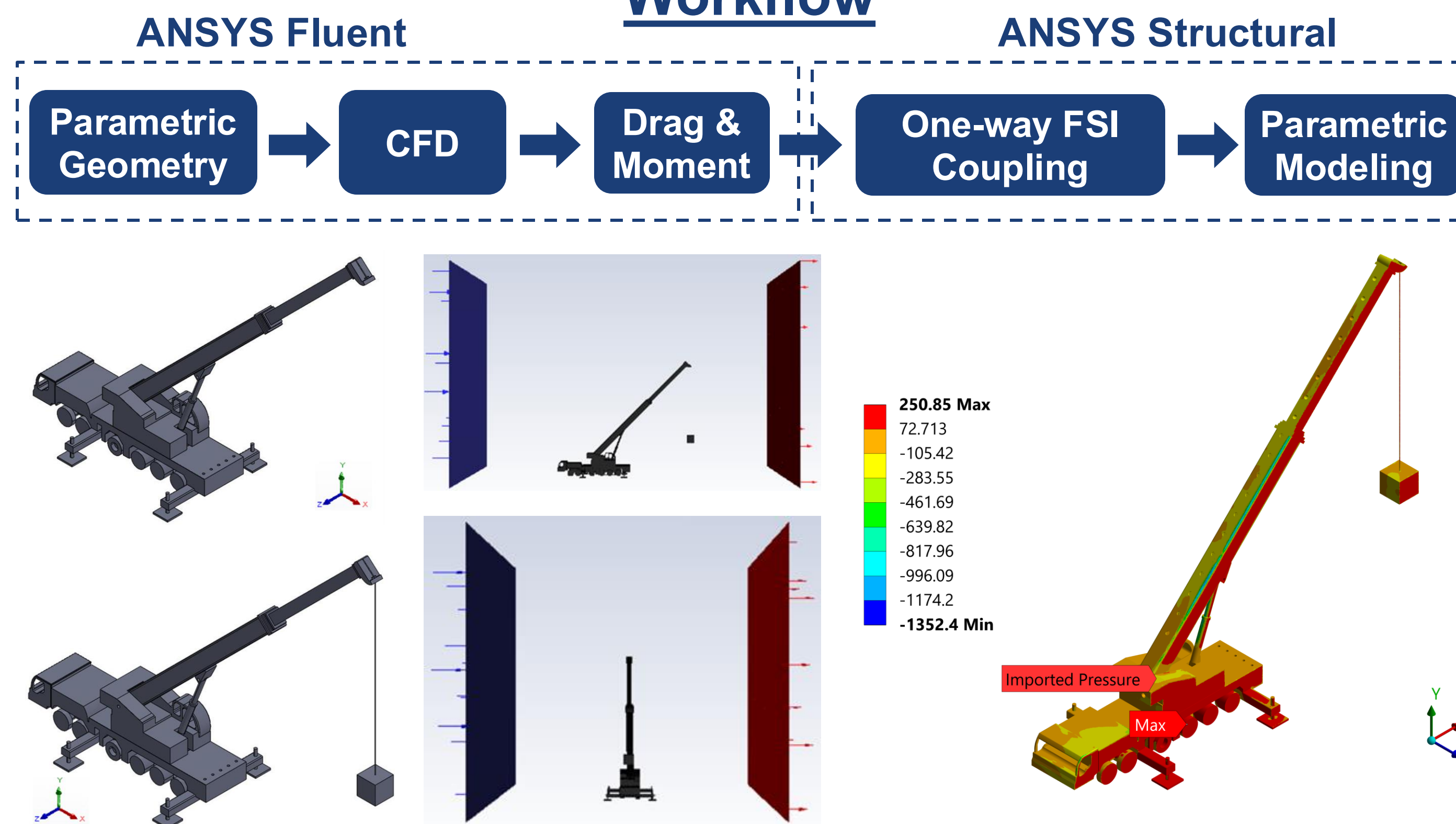
- Wind loading poses a safety risk for mobile cranes
- Payload swing in mobile crane amplifies wind effects.
- FSI study of mobile cranes help predict overturning moment and risk of toppling.



Project Objectives

- Develop CFD- FSI framework to evaluate mobile crane stability under wind loading.
- Capture deformation-dependent defects that are neglected by standards.
- Investigate how changing configurations(boom angle, boom length, wind direction, payload geometries) affect stability margins.

Method Workflow

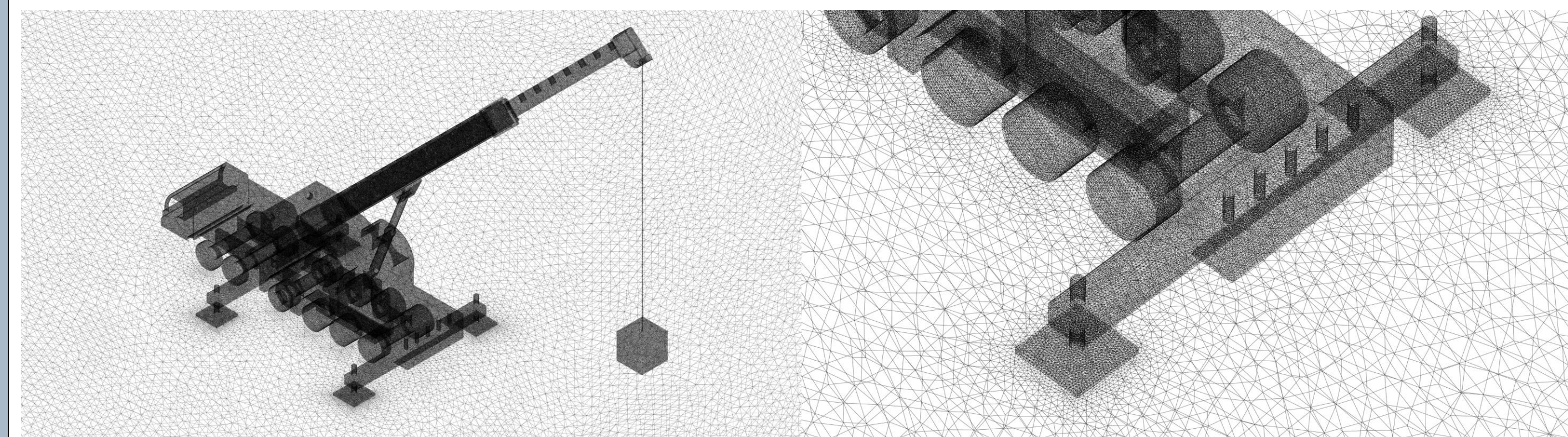


- Parametric geometry and load conditions with varying boom angle and boom lengths, wind velocity and wind angle.
- Steady RANS fluid simulations with fixed velocity inlet and pressure outlet.
- Converged static output is coupled with static structural to import fluid pressure on the crane structure.
- Structural effect(payload deflection, maximum stress, maximum strain) of the drag forces on mobile crane is evaluated.

Method – Meshing

Box enclosure was created around the crane to represent fluid domain. Meshing was performed as per parameters mentioned.

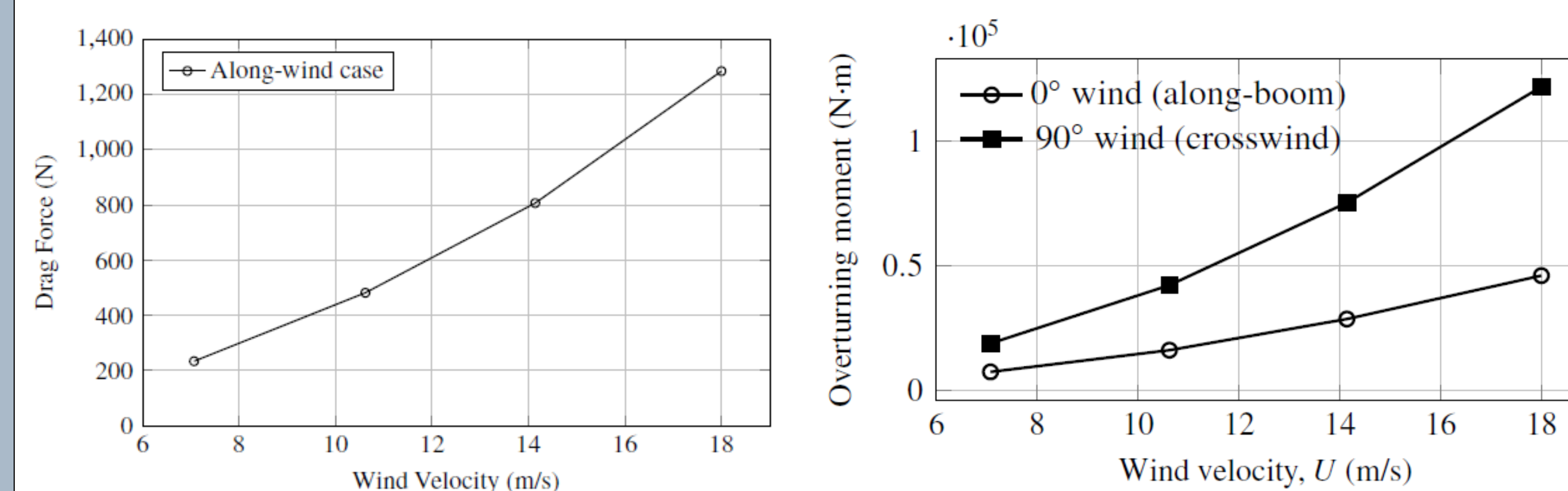
Parameter	Value / Setting
Meshing tool	ANSYS Meshing
Solver preference	ANSYS Fluent
Physics preference	CFD
Mesh type	Unstructured (hybrid)
Total nodes	5,025,319
Total elements	27,830,601
Enclosure (fluid domain) sizing	0.5 m (body sizing)
Crane & payload surface sizing	0.03 m (face sizing, 675 faces)



Mesh convergence study was performed to choose a computationally inexpensive but accurate mesh sizing for the fluent simulations.

Mesh	Cells	Drag (N)	Moment (Nm)	% Difference
Coarse	[376871]	[5477]	[36238]	–
Medium	[6717430]	[5597]	[38891]	[7%]
Fine	[27830601]	[5794]	[40078]	[3%]

Results – Drag Forces and Moments



Results –Payload Geometry and Wind angle

Velocity	0° Wind (Along-Body)					90° Wind (Crosswind)				
	$F_x (N)$	$F_y (N)$	$M_x (Nm)$	$M_y (Nm)$	$M_z (Nm)$	$F_z (N)$	$F_y (N)$	$M_x (Nm)$	$M_y (Nm)$	$M_z (Nm)$
18 m/s	5794	-518	-1206	2099	-40078	15780	4395	76631	-64467	-10375
18 m/s (sideward tipping)	–	–	–	–	–	–	–	121954	-64506	-10390
18 m/s (backward tipping)	–	–	-1559	658	-45994	–	–	–	–	–
14.14 m/s	3531	-228	-635	1101	-24300	9745	2721	47299	-39780	-7031
14.14 m/s (sideward tipping)	–	–	–	–	–	–	–	75320	-39771	-7028
14.14 m/s (backward tipping)	–	–	-799	434	-28537	–	–	–	–	–
10.62 m/s	2005	-155	-281	585	-13764	5548	1393	26881	-22587	-4370
10.62 m/s (sideward tipping)	–	–	–	–	–	–	–	42137	-22747	-4429
10.62 m/s (backward tipping)	–	–	-435	-43	-16003	–	–	–	–	–
7.07 m/s	886	-62	-275	393	-6233	2450.7	627	11923	-9999	-1858
7.07 m/s (sideward tipping)	–	–	–	–	–	–	–	18715	-10016	-1864
7.07 m/s (backward tipping)	–	–	-354	72	-7266	–	–	–	–	–

Actual velocity	0° wind direction					90° wind direction				
	Drag force _x (N)	Lift force (N)	Moment _x (Nm)	Moment _y (Nm)	Moment _z (Nm)	Drag force _z (N)	Lift force (N)	Moment _x (Nm)	Moment _y (Nm)	Moment _z (Nm)
18 m/s (222-ton payload)	7463	-544	-1857	3123	-46111	-18252	2026	-88717	102036	-11377
18 m/s (27-ton payload)	6402	-851	-1713	2377	-43073	-15792	2230	-79629	70173	-7316
18 m/s (No payload)	5959	-809	2604	-832	-43858	-14531	3437	-71623	56165	-3197
18 m/s (Spherical payload)	5794	-518	-1206	2099	-40078	-15780	4395	-76631	64467	-10375

Results – FSI

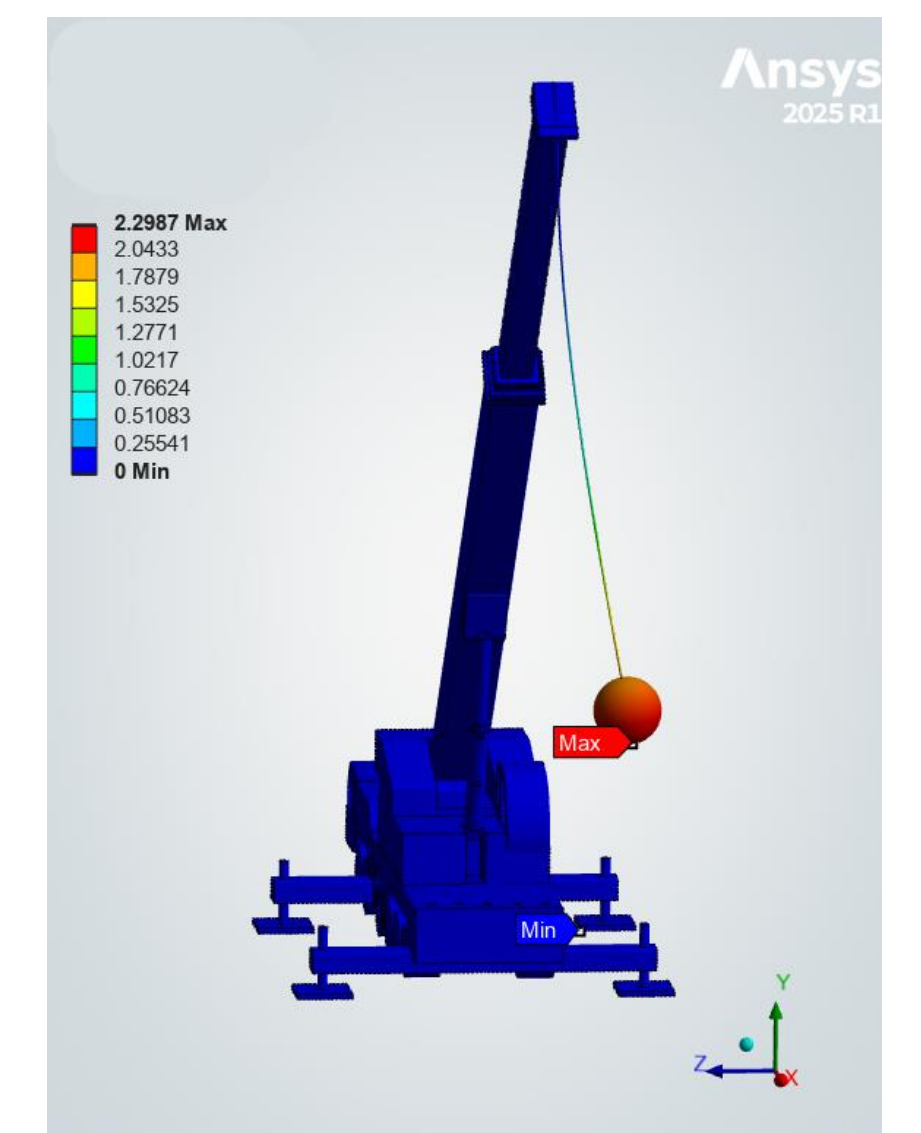
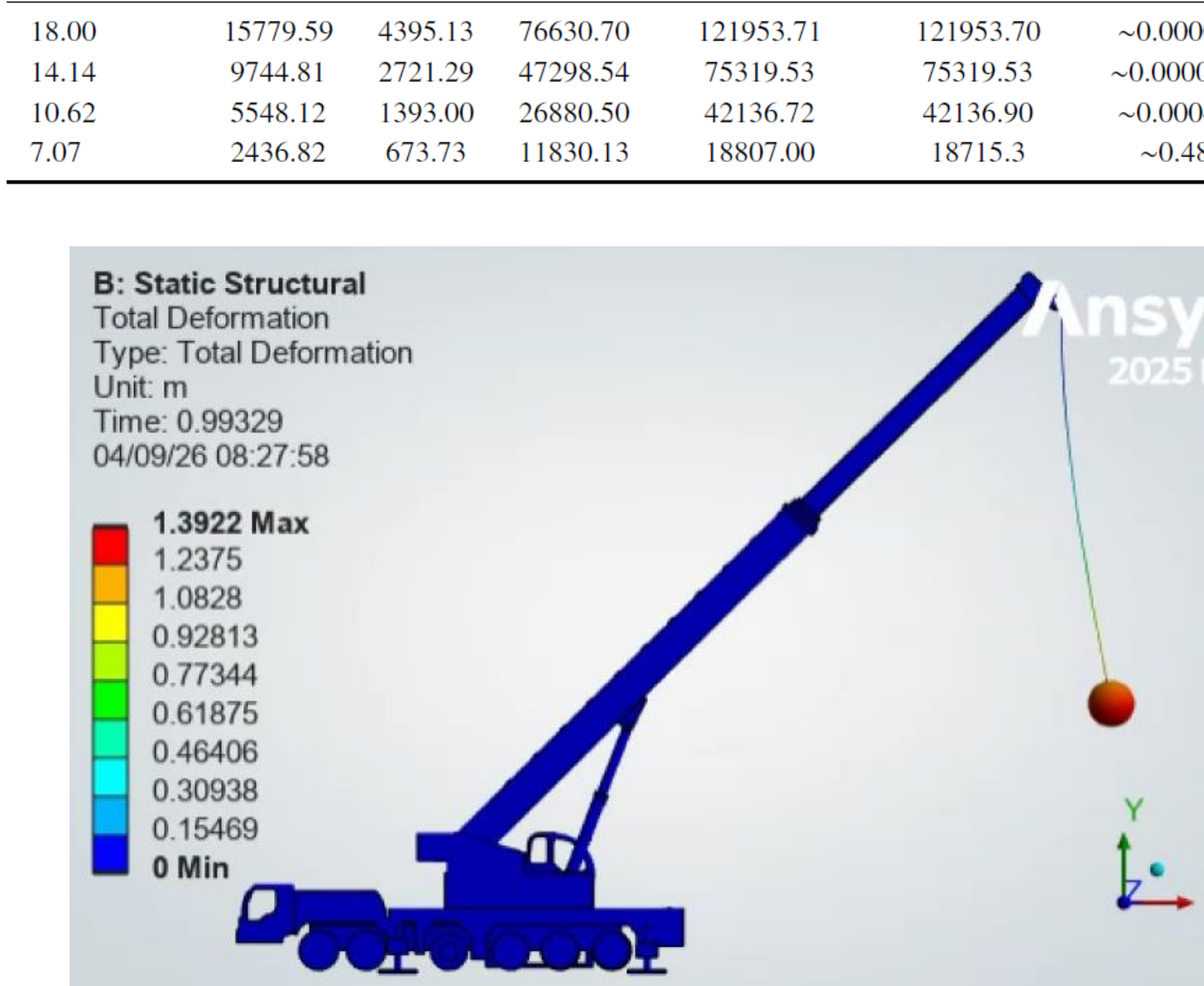
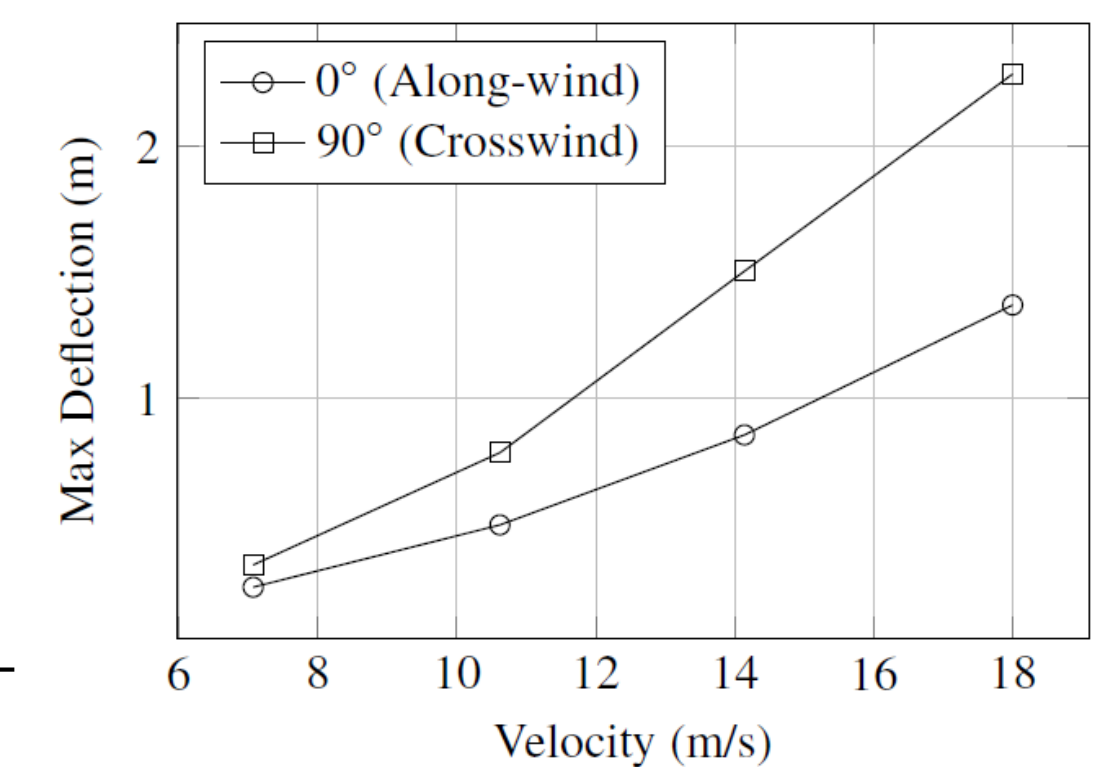
Wind-induced payload deflection shifts the center of mass, increasing the effective moment arm. This leads to amplified overturning moments compared to rigid-body assumptions.

$$\Delta M_{\text{geom}} \approx F_{\perp} \Delta r_{\perp}$$

$$\Delta M_{\text{geom}} \lesssim (886.2)(0.25) \approx 221.5 \text{ N} \cdot \text{m}.$$

$$\Delta M_{\text{geom}} \lesssim (2450.7)(0.34) \approx 833.24 \text{ N} \cdot \text{m}.$$

Wind speed (m/s)	F_z (N)	F_y (N)	$M_{x,O}$ (N-m)	Manual $M_{x,P}$ (N-m)	ANSYS $M_{x,P}$ (N-m)	% diff
18.00	15779.59	4395.13	76630.70	121953.71	121953.70	~0.00001%
14.14	9744.81	2721.29	47298.54	75319.53	75319.53	~0.000002%
10.62	5548.12	1393.00	26880.50	42136.72	42136.90	~0.00044%
7.07	2436.82	673.73	11830.13	18807.00	18715.3	~0.48%



Summary / Conclusions

- Overturning moment scales with U^2 wind velocity
- Crosswind loading produces 2–3 times higher moments than along-wind.
- Payload geometry matters: bluff bodies (cubic) → higher forces
- Deformation increases moment arm. amplifies instability

Acknowledgments

Support for the Crane Safety Research Center at Georgia Tech was provided by the Sarah Pantip Wong Family.

References

- [1] Timothy R Merinar and Paul H Moore. "NIOSH Alert: Preventing Worker Injuries and Deaths from Mobile Crane Tip-Over, Boom Collapse, and Uncontrolled Hoisted Loads". In: (2006).
- [2] Linda Levine. *Worker Safety in the Construction Industry: The Crane and Derrick Standards*. DIANE Publishing, 2011.
- [3] *Mobile and locomotive cranes: safety standard for cableways, cranes, derricks, hoists, hooks, jacks, and slings*. eng. ASME B30.5-2021. OCLC: 1380726725. New York, NY: American Society of Mechanical Engineers, 2021. isbn: 9780791874714.
- [4] *Cranes - Wind load assessment*. Tech. rep. ISO 4302:2016. Geneva, Switzerland: International Organization for Standardization, 2016.
- [5] Alan G Davenport and Jorge D Riera. *Wind effects on buildings and structures*. CRC Press, 1998.
- [6] Y. Tamura et al. "Wind load and wind-induced response estimations in the Recommendations for Loads on Buildings, AIJ 1993". In: *Engineering Structures* 18.6 (1996), pp. 399– 411. issn: 0141-0296. doi: [https://doi.org/10.1016/0141-0296\(95\)00121-2](https://doi.org/10.1016/0141-0296(95)00121-2).
- [7] Andreas Rauch et al. "Tip-Over Stability Analysis of Mobile Boom Cranes With Swinging Payloads". en. In: *Journal of Dynamic Systems, Measurement, and Control* 135.3 (May 2013), p. 031008. issn: 0022-0434, 1528-9028. doi: 10.1115/1.4023276. (Visited on 01/24/2026).
- [8] Daichi Fujioka. "Tip-over stability analysis for mobile boom cranes with single-and double-pendulum payloads". PhD thesis. Georgia Institute of Technology, 2010.